

Writing Assignment #1

The assignment should be a 1-2 page narrative. It should be typed, double spaced, 1" margins, font-size 12pts. A separate cover page is not required. Spelling, grammar and syntax should be appropriate to the subject. The assignment is due Wednesday, 27 October, 5:00 pm. The writing assignment will be returned with comments to you on Friday, 29 October, for revision. The revised version will be due Wednesday, 3 November, 5:00 pm.

The subject for the assignment is:

Show that the set functions

$$F = \{f : (0,1) \rightarrow \mathbb{R} \mid f \text{ is continuous on } (0,1)\}$$

with operation “addition” is a group. Furthermore, show that the subset B of F of bounded continuous functions on $(0,1)$, i.e., the subset

$$B = \{f \in F : \text{there exists } M_f > 0 \text{ for which } |f(x)| < M_f \text{ for all } x \in (0,1)\}$$

is a proper subgroup of F .